

RMFI handout

Operational Risk

Hull: Operational Risk, chapter 20

Operational risk:

the risk of loss or malfunction of people, internal processes and systems, or losses caused by external factors. It includes legal risk but does not include strategic and reputational risk.

Capital calculation methods:

- Basic Indicator Approach (BIA)
- Standardised Approach (STA)
- Alternative Standardised Approach (ASTA)
- Advanced Measurement Approach (AMA)

A consultation paper on regulatory change was prepared in 2016, based on which the 3 methods will be replaced by a risk-sensitive standard approach, the SMA (Standardized Measurement Approach). The introduction is constantly shifting, now 2025 is the target date.

Business lines Beta

Corporate finance 18%
Trade and sales 18%
Retail banking 12%
Commercial banking 15%
Payment and settlement 18%
Agency services 15%
Asset management 12%
Retail brokerage 12%

Type of oprisk events:

- external fraud
- internal fraud
- employment practices and workplace safety
- clients, products, and business practice
- damage to physical assets
- business disruption and system failures
- execution, delivery and process management

Exercises:

1. Find out the possible operational risks of TaketheMoney Bank!

TakeTheMoney Bank Plc. is a Hungarian bank that provides dynamic financial services featuring new technologies mainly for young customers. Customers are mostly from the younger generation, and they also use the services for their business and are more sensitive for speed and flexibility. The bank's server centre has moved into the basement of a Danube-side, central building. The IT systems and maintenance services are operated by an external, contracted partner. The bank is very proud of its 7/24, high quality internet-based services, which is in the focus of the company's reputation and advertising campaigns. TtM Bank is active on several social sites to communicate with young clients. The Bank, as a listed company, pays special attention to stock exchange reports. The current, most important task of management is an evaluation of the results of the most recent internal audit report. Internal audit urges anti-money laundering regulations, a code of ethics and reviewing customer information practices. The other priority is a recently disclosed corporate credit fraud, during which the collateral behind the loan was withdrawn and then the clients declared bankrupt. Bank security suspects that the clients involved had insider information.

2. A bank has classified its activities and determined the average gross income (\$ md) for the last three years for the activities:

Business Line	Gross Income (Mrd HUF)
Corporate Finance	5,5
Trade and Sales	2,8
Retail Banking	25,3
Commercial Banking	33,0
Payment and Settlement	0,5
Agency Services	0,8
Asset Management	14,4
Retail Brokerage	22,3

Retail Banking has a total loan portfolio of \$ 563 billion and Commercial Banking has a total loan portfolio of \$ 822 billion.

Calculate the bank's capital requirement due to operational risks using:

- the Basic Indicator Approach,
 - the Standard Approach and
 - the Alternative Standard Approach.
3. Model the operating losses due to computer system failure according to the Advanced Measurement Approach. Based on past experience, the number of risk events (failures) in a year follows a Poisson distribution with parameter $\lambda = 20$. The magnitude of the losses suffered for each failure follows a lognormal distribution with parameters $m = 10$ and $s = 3$.

- a) Use a simulation of 10,000 to determine the distribution of the annual loss (excluding the time value)! Plot the histogram of annual losses (between 0 and 50 million HUF, with 1 million aperture sizes)!
- b) What is the expected value of a single loss or an annual loss?
- c) What capital needs to be set aside for this operational risk element under Basel II. according to regulation (significance level: 99.9%)?
- d) Analyse the situation! Do you think it is worth taking out insurance?

Code

```
Sub oprisk_sim()
```

```
    Dim num As Double
```

```
    Dim lambda As Double
```

```
    Dim m As Double
```

```
    Dim s As Double
```

```
    Dim loss_counts() As Integer
```

```
    Dim yearly_losses() As Double
```

```
    Dim i As Double, j As Integer
```

```
    Range(Cells(1, 1), Cells(65536, 1)).Clear
```

```
    Application.ScreenUpdating = False
```

```
    num = Cells(1, 4)
```

```
    lambda = Cells(2, 4)
```

```
    m = Cells(3, 4)
```

```
    s = Cells(4, 4)
```

```
    ReDim loss_counts(1 To num)
```

```
    ReDim yearly_losses(1 To num)
```

```
    For i = 1 To num
```

```
        loss_counts(i) = poisson_rnd(lambda)
```

```
    Next i
```

```
    For i = 1 To num
```

```
        For j = 1 To loss_counts(i)
```

```
            yearly_losses(i) = yearly_losses(i) + Application.LogInv(Rnd, m, s)
```

Next j

Cells(i, 1) = yearly_losses(i)

Next i

Application.ScreenUpdating = True

End Sub

4. A bank's Business Indicator is 5,5 billion EUR. Over the past 10 years, there have been 8 operational risk incidents with the next loss amounts: 3, 7, 15, 65, 85, 150, 250 and 300 million EUR. How much is the bank's operational risk capital requirement under the SMA capital calculation method?